

Non-normal transient amplification, and how adaptation reduces it

2026-09-12. Companion to *Lyapunov_estimation_methods.md* §4.4 and to *fig_transient_amplification*. What was measured on the fast smoke (*data/topk_smoke_fast*), why the measure is taken on the dendritic block rather than the full Jacobian, what that implies, and how the reduction by adaptation could be quantified for the paper.

1. The measure

For a linear system $dz/dt = J z$, the spectral abscissa $\alpha(J) = \max \operatorname{Re} \lambda(J)$ gives the asymptotic growth rate, and the **numerical abscissa** $\omega(J) = \max \operatorname{eig}((J + J^T)/2)$ gives the largest possible *instantaneous* growth rate of the Euclidean norm: $d\|z\|/dt \leq \omega \|z\|$, with equality for the top eigenvector of the symmetric part. For a normal J (J^T commuting with J) the two coincide; for a non-normal J , $\omega > \alpha$, and the gap is the room for a perturbation to grow before the eigen-decay wins (Trefethen & Embree, *Spectra and Pseudospectra*). Every eigenvalue can be in the left half plane while ω is large and positive: that is transient amplification, and it is the mechanism Murphy & Miller call *balanced amplification* and Hennequin, Vogels & Gerstner exploit for movement generation in stable E/I networks.

The instantaneous Jacobian of the SRNN is evaluated along the trajectory (150 states per condition in the eig-heatmap stage), so ω and α are time series; *fig_transient_amplification* shows both, and the distribution of $\omega - \alpha$ per condition.

2. What was found (fast smoke, one seed, $n = 500$, medium of 40 states)

regime	$\omega(J_{xx})$ (1/s)	$\alpha(J_{xx})$ (1/s)	$\omega - \alpha$	λ_1
no adaptation	82 [79, 86]	7	≈ 82	+3.94
single-timescale SFA + STD	73 [69, 77]	6	≈ 73	+0.63
multiple-timescale SFA + STD	46 [44, 48]	0.3	≈ 46	-0.11

(median [IQR] over sampled states.) Two things stand out. First, the instantaneous growth bound is one to two orders of magnitude above the spectral abscissa in every regime: this is a strongly non-normal network, as a Dale's-law E/I network must be. Second, adaptation lowers ω from 82 to 46 s^{-1} , a 44% reduction, with the single-timescale regime in between.

Where the size comes from: $J_{xx} = (-I + W \operatorname{diag}(\theta'))/\tau_d$ with $\tau_d = 0.1$ s, so $\omega = (-1 + \lambda_{\max}(\operatorname{sym}(W \operatorname{diag} \theta')))/\tau_d$. A Dale's-law W has a rank-one mean structure (every E column positive, every I column negative) whose largest *singular* value grows like $\sqrt{n} \cdot \mu$ while its outlier *eigenvalues* stay near the block means; the symmetric part inherits the singular value, so $\lambda_{\max}(\operatorname{sym} W_{\text{eff}}) \approx 5\text{--}9$ and $\omega \approx 40\text{--}80$ s^{-1} . The direction that achieves it is the E-versus-I difference mode: a perturbation that raises E and lowers I is amplified through the loop before the balanced sum mode decays. That is Murphy & Miller's balanced amplification, and it is physical, not an artefact.

3. Why the dendritic block J_{xx} , not the full Jacobian

The first implementation took ω on the full $N \times N$ Jacobian ($N = 4000$: x, three SFA variables and eight STD variables per neuron). It read 50–180 s^{-1} and ranked the regimes plausibly, but the number is not trustworthy, for a reason that is structural rather than numerical:

The numerical abscissa is not invariant to a diagonal rescaling of the state. If $z = D y$ with D diagonal, J becomes $D^{-1} J D$, whose eigenvalues (hence α) are unchanged, but whose symmetric part is not: ω depends on the units in which each state variable is measured. The full Jacobian mixes rows in units of $1/\tau_d = 10$ s^{-1} (x), $1/\tau_a = 0.1\text{--}4$ s^{-1} (SFA) and $1/\tau_{\text{rec}} + r/\tau_{\text{rel}} \approx 0.3\text{--}5$ s^{-1} (STD), and its off-diagonal couplings are asymmetric by construction: $J(x, a) = -c_{\text{eff}} W \operatorname{diag}(\varphi')/\tau_d$ against $J(a, x) = \operatorname{diag}(\varphi')/\tau_a$, differing by a factor of order $W/(c \tau_d) \cdot \tau_a$. The symmetric part of that pair is dominated by whichever block happens to have the larger units, so ω on the full J measures how we chose to write the state vector, not how the dynamics amplify. There is no canonical normalisation for a state whose components are a potential, a dimensionless adaptation current and a synaptic resource.

The dendritic block J_{xx} has one unit throughout, and it is exactly the object Hennequin et al. and Murphy & Miller analyse: a rate network with effective connectivity $W_{\text{eff}} = W \text{diag}(\theta')$, where $\theta' = \varphi'(x_{\text{eff}}) \cdot \Pi b \cdot \Pi g$ is the slope of the synaptic output at the operating point. So the restriction is what makes the number comparable to the literature.

What the restriction costs. J_{xx} freezes the adaptation and depression states. Its ω therefore captures how adaptation changes the *operating point* (the slope φ' where each neuron sits, and the depression factor Πb multiplying each synapse) but not adaptation's *dynamic* negative feedback on the timescale of the transient. The $82 \rightarrow 46 \text{ s}^{-1}$ reduction is the static part: multiple-timescale STD depresses the effective synaptic gain and SFA moves neurons to shallower parts of the nonlinearity, and together they shrink the singular values of W_{eff} . The dynamic part (the adaptation current rising during a transient and cutting it short) is invisible to this measure, and it is the part that operates on the 0.25 s to 10 s timescales where a transient of a few τ_d would be most affected. So the figure *understates* adaptation's effect on transient amplification, in a direction that makes the claim conservative.

4. Implications

- The three regimes differ in the *bound* on transient growth by a factor of ~ 2 , on top of differing in the asymptotic sign. “Stable but supports transient divergence” (the Introduction’s claim) has a number attached: in the multiple-timescale regime $\alpha \approx 0.3 \text{ s}^{-1}$ and $\lambda_1 \approx -0.11 \text{ s}^{-1}$, yet a well-chosen dendritic perturbation can grow at up to 46 s^{-1} for a fraction of τ_d .
- The bound is achieved by the E/I difference mode. Whether the network’s own fluctuations actually excite that mode is a separate question, and is what the trajectory-side measures answer (the fraction of time the local Lyapunov rate is positive; the 0.2 s finite-time exponent; the positive-excursion length, all stored per sweep job).
- Because ω scales with the singular values of W_{eff} , it will track the synaptic-gain and μ sweeps almost by construction. The informative comparison is *across adaptation regimes at fixed W* , which is how the stage samples (same seed for all conditions).

5. How to quantify adaptation’s reduction, for the paper

In increasing order of effort and completeness:

1. **$\omega - \alpha$ on J_{xx} , per condition, paired by network** (done). Report the median over states and seeds and the paired reduction relative to no adaptation. Cheap; static; understates the effect.
2. **A scale-free non-normality index** on the same block: the ratio of the largest singular value to the spectral radius of W_{eff} , $\sigma_{\text{max}}(W_{\text{eff}})/\rho(W_{\text{eff}})$, or Henrici’s departure from normality $\|W_{\text{eff}}\|_F^2 - \sum |\lambda_i|^2$. These separate “the gain went down” from “the matrix became more normal”, which ω alone conflates.
3. **Maximal transient gain with adaptation active:** $G(t) = \|P_x \exp(J t) P_x^T\|_2$, the largest growth of a *dendritic* perturbation into a *dendritic* response through the full system over time t , with the adaptation and depression states free to respond. Because the input and output are both in x , this is invariant to the state scaling that spoils ω on the full J , and it contains the dynamic feedback that J_{xx} omits. Compute by propagating the n -column basis $[I_n; 0]$ through the tangent equation with the matrix-free jacobian_times for t up to $\sim 1 \text{ s}$ (about 20 s per sampled state at $N = 4000$; 20 states per condition is affordable) and taking the top singular value of the x -rows. The peak of $G(t)$ over t , and the time of the peak, per condition, is the number I would put in the paper: “a dendritic perturbation is amplified at most G_{max} -fold, at t_{peak} , before adaptation reverses it”.
4. **Hennequin’s evoked energy**, the integral of $\|\Delta r\|^2$ over the response to an optimal initial condition, obtained from a Lyapunov equation for a *stable* linearisation. Cleanest for a fixed stable J_{xx} ($n = 500$, so the Lyapunov solve is milliseconds), but the frozen block is not stable in every regime ($\alpha > 0$ in two of three), and on the full J it inherits the scaling problem unless restricted to x -in/ x -out as in item 3.

Items 1 and 2 are a few lines in `run_eig_heatmap`’s `sample_eigenvalues`; item 3 is a small stage of its own and is the one that answers the question as posed (“how much does adaptation reduce transient amplification”) with adaptation actually acting.

6. Results: transient gain with adaptation frozen vs active (2026-09-13)

Item 3 is built: `run_transient_gain` (stage `transient_gain`, after `lyapunov_spectrum`), `fig_transient_gain`, `fig_transient_gain_excursions`, on the statics `SRNNCellTypePairs.transient_gain`, `leading_direction_at`

and excursion_samples. The stage propagates the n dendritic unit vectors through the tangent equation from sampled states of the noisy trajectory and reads the x-in / x-out block every 20 ms, with THREE propagators: frozen_x (J_{xx} fixed at the state), frozen_full (the full J fixed at the state) and active ($J(S(t))$ along the trajectory). Verified against expm on small networks to $2e-6$ (test_transient_gain).

6.1 Fast smoke, $n = 500$, one seed, $T = 20$ s, horizon 1 s

Six regular states per regime (data/transient_gain_smoke, figs/transient_gain_smoke); $G_{\max}$ over the 1-s horizon, median [min, max]:

regime	λ_1	J_{xx} frozen	J frozen	active	active t_{peak}
no adaptation	+4.0	47 000 [11 000, 540 000]	same	1 200 [370, 4 700]	1.0 (horizon)
single-timescale	+0.44	11 000 [360, 89 000]	143 [9, 520]	30 [19, 69]	0.87 [0.24, 1.0]
multiple-timescale	-0.11	39 [12, 1 200]	6.7 [5.3, 54]	8.6 [5.0, 19]	0.18 [0.14, 1.0]

What it says:

- **Adaptation's dynamic feedback is large.** Freezing the full J instead of only J_{xx} cuts the 1-s gain by $\sim 80\times$ (single-timescale) and $\sim 6\times$ (multiple-timescale). This is the part the $\omega - \alpha$ figure cannot show. In the no-adaptation regime the two frozen propagators are identical ($N = n$), which is the built-in sanity check.
- **The frozen gain in an unstable regime is not a transient.** Where $\alpha(J_{xx}) > 0$ (no and single-timescale adaptation, $\alpha \approx +7$ and $+6$) the frozen curve is $e^{\{\alpha t\}}$ times a non-normal prefactor and grows without bound, so its "peak" is the horizon. The comparison that means something there is the gain at a fixed t , or the active vs frozen curves at the same t ; only the multiple-timescale regime has a genuine peak (0.18 s) and decay.
- **The active gain is the honest number**, and it tracks the regime's λ_1 : the no-adaptation trajectory amplifies $\sim e^{\{7\}}$ in a second along its worst direction ($\lambda_1 = +4$ plus non-normal boost), single-timescale ~ 30 , multiple-timescale ~ 9 with a peak at 0.18 s and recovery.
- **Active vs frozen-full is not one-signed.** At $n = 500$ the active gain is below the frozen-full gain in the two unstable regimes and slightly ABOVE it in the stable one (8.6 vs 6.7); on the 40- and 60-neuron test networks the active gain also came out above the frozen ones. The drift of the state through regions of stronger local expansion (the local rate fluctuates around λ_1) can add more than the frozen picture allows. So the frozen propagators are not bounds; report all three.
- **Directions.** The worst-case direction at the peak has a participation ratio of ~ 60 of 500 neurons, E fraction ~ 0.5 , and $|\cos| \leq 0.15$ with the leading Lyapunov direction: the direction that amplifies most over a second is NOT the direction the dynamics are expanding along. The E/I difference mode is the best of the named directions in every regime (the balanced-amplification prediction holds), the E/I sum mode is the worst (below 1 in the adapted regimes: a uniform push is damped), and the noise-average gain is near 1 in the adapted regimes – adaptation makes the network's own additive noise see no amplification on average while a worst-case perturbation still gains 10-30 $\times$.

6.2 Excursion onsets vs quiet states (Next 2)

In 20 s the fast smoke found 2 onsets and 0 quies in the single-timescale regime and 2 onsets, 1 quiet in the multiple-timescale one – too few for the contrast; the medium mode ($T = 40$ s, 2 seeds, up to 8 of each) is the first real test. What the few samples show: the onset states' active $G_{\max}$ (33 and 13) sits above the regular-state medians (30 and 8.6), and the optimal direction at onset has $|\cos| 0.02-0.16$ with the leading Lyapunov direction. If that alignment stays low with more samples, the excursions are not "the network recruiting its own non-normal mode" in the worst-case sense; they may still be amplification along a different, lower-gain direction, which the G_{lyap} reading measures directly.

6.3 Where an intermittent-but-stable network is (for Next 3)

scripts/examples/find_intermittent_stable.m queries the medium sweep for jobs with $\lambda_1 < 0$, the leading local rate positive $\geq 25\%$ of the time and mean excursions ≥ 0.1 s: 78 / 78 / 114 of ~ 1 200-1 500 successful jobs per regime. The strongest candidates (p95 of the 0.2-s finite-time exponent ≈ 10 -13 s^{-1} , $\lambda_1 \approx -0.01$ to -0.1) are the multiple-timescale regime with $\mu_{IE_relative}$ 2.06 or $\mu_{EE_relative}$ 13.4-15.7 in the 1-D sensitivity sweeps (several reps each), and the single-timescale regime at $n = 190$. Those are the networks to build the stimulation-direction analysis on.

References

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